



# Risk-Averse Humanitarian Logistics

A Mean-CVaR Approach to Disaster Relief Networks

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# Today's plan

## Part I. Problem and Background

1. Motivation and Humanitarian Logistics
2. Research Objective
3. Literature Review
4. Uncertainty in Humanitarian Logistics

## Part II. Model and Case Study

5. CVaR and Risk Measures
6. Risk-Neutral vs. Risk-Averse
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# 1. Motivation and Humanitarian Logistics

- ▶ After a disaster people need water, food and medicine within hours.
- ▶ Aid is moved through relief hubs and roads that may themselves be damaged.
- ▶ Strategic choices such as **where to build hubs** and **what to stock** are made *before* the event.
- ▶ Decisions must survive rare but catastrophic scenarios, not just the average day.

## Central challenge

Demand, transport capacity and usable inventory are all uncertain when the plan is fixed. Wrong plans cost lives.

## 2. Research Objective

### Research question

How does moving from a risk-neutral to a CVaR-based risk-averse plan change **unmet demand** and **response time** in disaster relief planning?

### What we do:

- ▶ Build a two-stage stochastic MILP with truck and air modes.
- ▶ Solve it in Python and Gurobi on the Rawls Gulf Coast case study (15 real historical hurricanes).
- ▶ Compare a risk-neutral baseline against a calibrated risk-averse policy.
- ▶ Trace the trade-off between expected cost and tail unmet demand.

### 3. Literature Review

| Paper                    | Main idea                                                                | What we take                                                       |
|--------------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------|
| Rawls & Turnquist (2010) | Two-stage stochastic MIP for supply pre-positioning under hurricane risk | Network, first- and second-stage decisions, 15 hurricane scenarios |
| Yi & Özdamar (2007)      | Multi-period integrated location and distribution model                  | Period-indexed flows, dynamic demand over the horizon              |
| Noyan (2012)             | Mean-CVaR two-stage stochastic programming for disaster preparedness     | Risk-averse objective and linearised CVaR                          |

#### Research scope

We focus on **mean-CVaR stochastic programming**. Distributionally robust reformulations sit outside our seminar scope.

## 4. Uncertainty in Humanitarian Logistics

### What is uncertain:

- ▶ **Demand.** How many people need water, food, medicine, and where.
- ▶ **Transport.** Which roads and airlifts stay usable.
- ▶ **Inventory.** How much pre-positioned stock survives at each hub.

### Why deterministic optimisation fails:

- ▶ A single “expected demand” input hides the tail.
- ▶ Optimising for the average leaves rare but severe events unprotected.

### Modelling choice

Represent uncertainty by a **finite set of scenarios**  $s \in S$  with probabilities  $p_s$ . Each scenario carries its own demand, its own damaged links, and its own surviving stock.

## 5. CVaR and Risk Measures

**Value-at-Risk (VaR)** at level  $\alpha$ :

$$\text{VaR}_\alpha(L) = \inf\{\eta : \mathbb{P}(L > \eta) \leq 1 - \alpha\} \quad (1)$$

In 95% of scenarios the loss stays below VaR. Silent about how bad the worst 5% can be.

**Conditional Value-at-Risk (CVaR):**

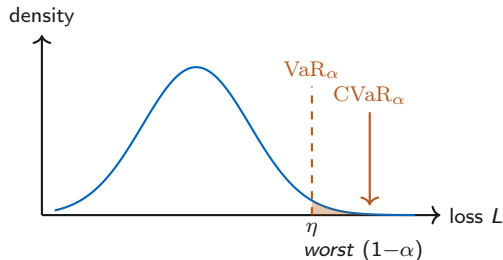
$$\text{CVaR}_\alpha(L) = \mathbb{E}[L \mid L \geq \text{VaR}_\alpha(L)] \quad (2)$$

Average loss in the worst  $(1 - \alpha)$  tail. Coherent, tail-aware, linearisable.

**Rockafellar and Uryasev linearisation:**

$$\text{CVaR}_\alpha(L) = \min_{\eta, \xi_s} \left\{ \eta + \frac{1}{1-\alpha} \sum_s p_s \xi_s \right\} \quad (3)$$

with  $\xi_s \geq L_s - \eta$  and  $\xi_s \geq 0$ .



*VaR is a threshold. CVaR is the average of losses beyond it.*

## 6. Risk-Neutral vs. Risk-Averse Planning

**Toy example.** Two policies A and B facing three equally likely scenarios.

|                         | Unmet demand per scenario |        |            | Expected mean | CVaR <sub>0.67</sub> tail |
|-------------------------|---------------------------|--------|------------|---------------|---------------------------|
|                         | Mild                      | Medium | Extreme    |               |                           |
| Policy A (risk-neutral) | 0                         | 20     | <b>200</b> | 73.3          | <b>200</b>                |
| Policy B (risk-averse)  | 30                        | 40     | 80         | 50.0          | <b>80</b>                 |

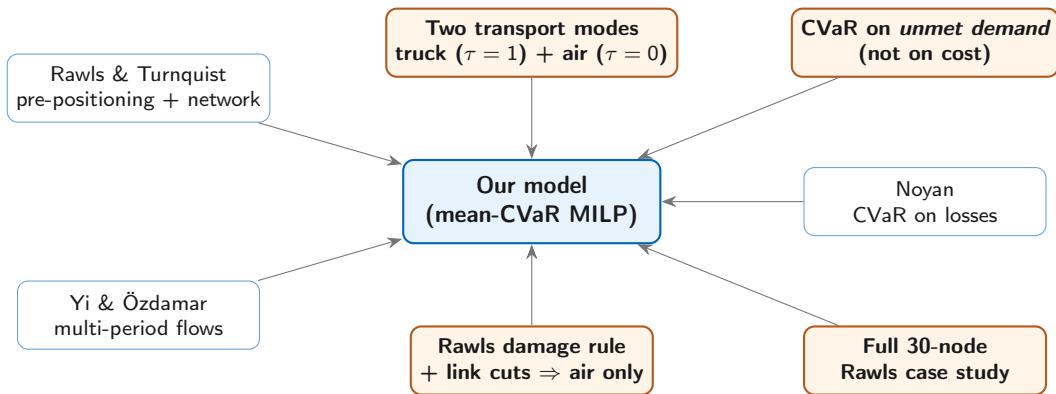
- ▶ A wins on the average. B wins in the worst case.
- ▶ CVaR shifts optimisation from “how does it usually look” to “how bad is the bad case”.
- ▶ Our model contains both. The weight  $\lambda$  decides how much tail matters.

### Why CVaR for humanitarian logistics

A relief planner cares *how bad* the worst hurricanes can be, not just whether they cross a threshold. CVaR answers exactly that question and can be embedded in a linear programme.



## 7. Our Contribution



## 8. Mathematical Model (1/5) Sets and indices

| Symbol                | Description                                              |
|-----------------------|----------------------------------------------------------|
| $N$                   | Set of all nodes in the relief network.                  |
| $I \subseteq N$       | Set of candidate facility locations.                     |
| $D \subseteq N$       | Set of demand locations.                                 |
| $A$                   | Set of directed transportation arcs $(i, j)$ .           |
| $K$                   | Set of relief commodities (water, food, medicine).       |
| $T = \{1, \dots, H\}$ | Set of operational time periods in the response horizon. |
| $T_0 = \{0\} \cup T$  | Set of inventory time periods, including pre-disaster.   |
| $S$                   | Set of disaster scenarios.                               |
| $L$                   | Set of facility size options.                            |
| $M$                   | Set of transportation modes (truck, air).                |

### Reading

### tip

Every quantity below carries the scenario index  $s$ . That is what makes the model stochastic.

## 8. Mathematical Model (2/5) Parameters

| Symbol         | Description                                                                                                      |
|----------------|------------------------------------------------------------------------------------------------------------------|
| $p_s$          | Probability of scenario $s$ , with $\sum_{s \in S} p_s = 1$ .                                                    |
| $d_{ikts}$     | Demand for commodity $k$ at node $i$ in period $t$ under scenario $s$ ( $d_{ikts} = 0$ if $i \notin D$ ).        |
| $\gamma_{iks}$ | Fraction of stock of commodity $k$ at hub $i$ that survives under scenario $s$ ( $0 \leq \gamma_{iks} \leq 1$ ). |
| $U_{ijmts}$    | Transportation capacity on arc $(i, j)$ by mode $m$ in period $t$ under scenario $s$ .                           |
| $F_{il}$       | Fixed cost of opening a facility of size $l$ at hub $i$ .                                                        |
| $M_l$          | Storage capacity of a facility of size $l$ .                                                                     |
| $b_k$          | Storage space required per unit of commodity $k$ .                                                               |
| $q_k$          | Acquisition cost per unit of commodity $k$ .                                                                     |
| $c_{ijkmts}$   | Unit transportation cost for commodity $k$ on arc $(i, j)$ , mode $m$ , period $t$ , scenario $s$ .              |
| $e_k$          | Capacity consumption per unit of commodity $k$ on transportation arcs.                                           |
| $h_k$          | Holding cost per unit of remaining inventory of commodity $k$ .                                                  |
| $\rho_k$       | Penalty cost per unit of unmet demand of commodity $k$ .                                                         |
| $\tau_m$       | Delivery lead time (in periods) for mode $m$ . $\tau_{\text{truck}} = 1$ , $\tau_{\text{air}} = 0$ .             |
| $\alpha$       | CVaR confidence level, $0 < \alpha < 1$ . We use $\alpha = 0.95$ .                                               |
| $\lambda$      | Risk-aversion weight, $\lambda \geq 0$ .                                                                         |

## 8. Mathematical Model (3/5) Decision variables

| Variable                         | Description                                                                          |
|----------------------------------|--------------------------------------------------------------------------------------|
| $y_{il}$                         | Binary variable, 1 if a facility of size $l$ is opened at node $i$ , 0 otherwise.    |
| $r_{ik}$                         | Amount of commodity $k$ pre-positioned at candidate hub $i$ .                        |
| $x_{ijkmts}$                     | Flow of commodity $k$ from $i$ to $j$ by mode $m$ in period $t$ under scenario $s$ . |
| $l_{ikts}$                       | Inventory of commodity $k$ at node $i$ at the end of period $t$ under scenario $s$ . |
| $w_{ikts}$                       | Unmet demand of commodity $k$ at node $i$ in period $t$ under scenario $s$ .         |
| <b>Additional risk variables</b> |                                                                                      |
| $\eta$                           | Value-at-Risk threshold for total unmet demand across scenarios.                     |
| $\xi_s$                          | Excess unmet demand above the VaR threshold in scenario $s$ .                        |
| $Z_s$                            | Realised first- plus second-stage cost under scenario $s$ .                          |
| $S_s$                            | Total unmet demand summed across nodes, commodities and periods under $s$ .          |

## 8. Mathematical Model (4/5) Objective and key constraints

### Objective function

$$\min \underbrace{\sum_{s \in S} p_s Z_s}_{\text{expected cost}} + \lambda \underbrace{\left[ \eta + \frac{1}{1-\alpha} \sum_{s \in S} p_s \xi_s \right]}_{\text{CVaR}_{\alpha} \text{ of unmet demand}}$$

with  $Z_s = C^{\text{first}} + Q_s$ ,  $S_s = \sum_{t, i \in D, k} w_{ikts}$  and  $\xi_s \geq S_s - \eta$ .

### Key constraints

- ▶ Multi-mode inventory balance with delivery delays  $\tau_m$ .
- ▶ Hub capacity  $\sum_k b_k r_{ik} \leq \sum_l M_l y_{il}$  and at most one size per hub.
- ▶ Initial usable stock  $I_{ik0s} = \gamma_{iks} r_{ik}$  (damage rule).
- ▶ Mode-specific transport capacity  $\sum_k e_k x_{ijkmts} \leq U_{ijmts}$ .
- ▶ Shortage bound  $0 \leq w_{ikts} \leq d_{ikts}$  at demand nodes.

### Next slide

The next slide stacks the full formulation with every equation numbered in reading order.

## 8. Mathematical Model (5/5) Complete risk-averse MILP

$$\min \sum_{s \in S} p_s Z_s + \lambda \left[ \eta + \frac{1}{1-\alpha} \sum_{s \in S} p_s \xi_s \right] \quad (4)$$

$$\text{s.t. } Z_s = C^{\text{first}} + Q_s, \quad \forall s \in S, \quad (5)$$

$$C^{\text{first}} = \sum_{i,l} F_{il} y_{il} + \sum_{i,k} q_k r_{ik}, \quad (6)$$

$$Q_s = \sum_{t,(i,j),k,m} c_{ijkmts} x_{ijkmts} + \sum_{t,i,k} h_k l_{ikts} + \sum_{t,i \in D,k} p_k w_{ikts}, \quad \forall s, \quad (7)$$

$$S_s = \sum_{t,i \in D,k} w_{ikts}, \quad \forall s \in S, \quad (8)$$

$$\sum_k b_k r_{ik} \leq \sum_l M_l y_{il}, \quad \sum_l y_{il} \leq 1, \quad \forall i \in I, \quad (9)$$

$$l_{ik0s} = \gamma_{iks} r_{ik}, \quad \forall i \in I, k, s, \quad (10)$$

$$l_{ikts} = l_{ik,t-1,s} + \sum_{j,m} x_{jikm,t-\tau_m,s} - \sum_{j,m} x_{ijkmts} - d_{ikts} + w_{ikts}, \quad \forall i, k, t, s, \quad (11)$$

$$\sum_k e_k x_{ijkmts} \leq U_{ijmts}, \quad \forall (i,j) \in A, m, t, s, \quad (12)$$

$$0 \leq w_{ikts} \leq d_{ikts}, \quad \forall i \in D, k, t, s, \quad (13)$$

$$\xi_s \geq S_s - \eta, \quad \xi_s \geq 0, \quad \forall s \in S, \quad (14)$$

$$y_{il} \in \{0, 1\}, \quad r_{ik}, x_{ijkmts}, l_{ikts}, w_{ikts} \geq 0, \quad \eta \in \mathbb{R}. \quad (15)$$

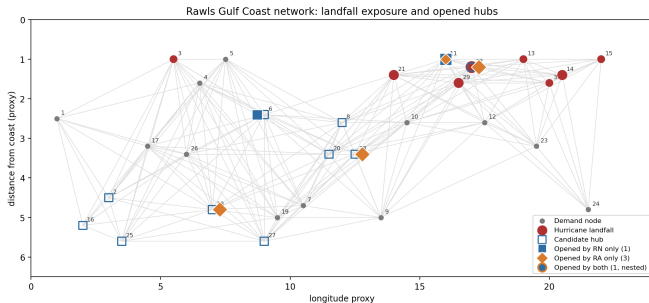
### Reading

(4) mean-CVaR objective. (11) multi-mode inventory balance with delivery delays  $\tau_m$ . (14) Rockafellar-Uryasev linearisation. Setting  $\lambda = 0$  recovers the risk-neutral model.

## 9. Case Study and Data

### Rawls & Turnquist (2010) Gulf Coast:

- ▶ 30 nodes on a coastal-to-inland layout.
- ▶ 11 candidate hubs (coastal + inland).
- ▶ 15 real hurricanes, categories 1 to 5 from the source paper.
- ▶ Category  $\geq 3$  destroys landfall stock ( $\gamma = 0$ ), category  $\leq 2$  leaves 50% ( $\gamma = 0.5$ ).
- ▶ Unusable roads cut truck capacity to zero. Air stays open.



### Reconstructed vs. verbatim from Rawls

**Verbatim.** Category, landfall, unusable links, per-commodity demand, scenario probabilities.

**Reconstructed and documented.** Node coordinates, exact arc list, multi-period split (0.50, 0.30, 0.20), air-mode data.

## 10. Experimental Design (data $\rightarrow$ code $\rightarrow$ results)



### Experimental setup

- ▶ **Baseline policy**
  - ▶ Risk-neutral ( $\lambda = 0$ ).
- ▶ **Sensitivity analysis**
  - ▶ Evaluate  $\lambda \in \{0, 5, 10, 15, 18, 22\}$ .
  - ▶ Analyse the cost vs. risk trade-off.
- ▶ **Representative risk-averse policy**
  - ▶  $\lambda = 18$ ,  $\alpha = 0.95$ .
  - ▶ Selected from the sensitivity analysis for detailed comparison.

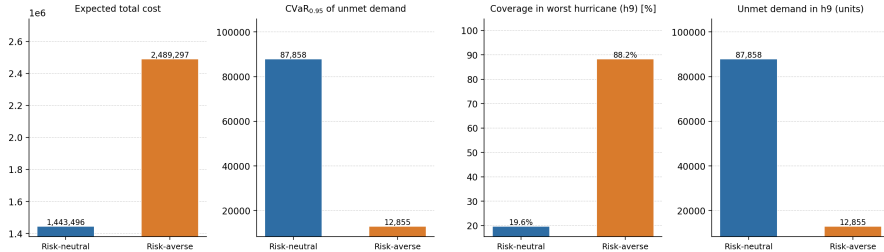
### Performance measures

- ▶ **Efficiency**
  - ▶ Expected cost  $\mathbb{E}[Z]$ .
  - ▶ Solve time (about 11 to 14 s per  $\lambda$ ).
- ▶ **Risk**
  - ▶  $\text{CVaR}_{0.95}$  of unmet demand.
  - ▶ Empirical CDF of unmet demand.
- ▶ **Service**
  - ▶ Coverage by scenario, period, commodity.
- ▶ **Network**
  - ▶ Number of hubs opened.



# 11. Results (headline)

Policy comparison ( $\lambda = 0$  vs  $\lambda = 18$ ) — Rawls Gulf Coast



| Metric                            | Risk-neutral | Risk-averse |
|-----------------------------------|--------------|-------------|
| $\mathbb{E}[Z]$                   | 1,443,496    | 2,489,297   |
| CVaR <sub>0.95</sub> unmet demand | 87,858       | 12,855      |
| Expected coverage                 | 83.5%        | 93.4%       |
| Coverage in h9 (Cat. 5)           | 19.6%        | 88.2%       |
| Period-1 coverage in h9           | 0.9%         | 76.5%       |
| Hubs opened                       | 2            | 4           |
| Solve time (s)                    | 10.8         | 13.4        |

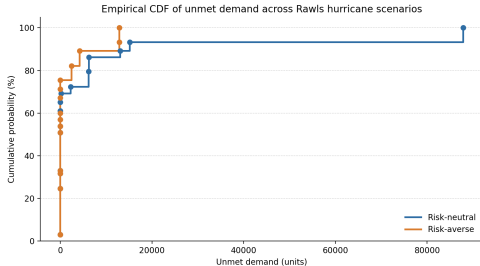
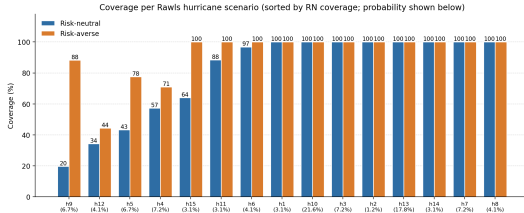
## Headline

Cost. +72%

Tail unmet demand. -85%

Worst-hurricane coverage.  
19.6% → 88.2%

## 12. Discussion



### Where the gap actually is:

- ▶ Coverage is essentially 100% on high-probability hurricanes for both policies.
- ▶ The risk-averse policy closes the gap in the tail (h9, h12, h15, h5, h4).

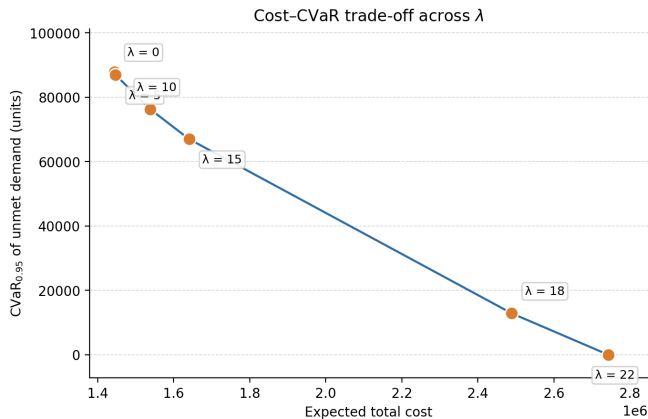
### CDF of unmet demand:

- ▶ The risk-averse CDF rises faster and saturates at a much smaller shortage.
- ▶ That shape is exactly the CVaR improvement in the headline table.

### Per-commodity view (worst hurricane):

- ▶ Water is fine under both policies.
- ▶ Food coverage 13% → 100%.
- ▶ Medicine coverage 20% → 86%.

## 12b. Pareto trade-off across $\lambda$



Exact values

| $\lambda$ | $E[Z]$    | $CVaR(S)$ |
|-----------|-----------|-----------|
| 0         | 1,443,496 | 87,858    |
| 5         | 1,447,014 | 86,933    |
| 10        | 1,538,330 | 76,250    |
| 15        | 1,640,377 | 66,993    |
| 18        | 2,489,297 | 12,855    |
| 22        | 2,743,204 | 0         |

Shallow regime  $\lambda \leq 15$  trims a few thousand units of tail unmet demand for a small cost bump. Sharper regime  $15 < \lambda \leq 22$  opens an extra hub and nearly eliminates the tail.

## 13. Managerial Insights

1. **Tail coverage has a price tag.** A calibrated risk-averse policy raises expected cost by 72% but cuts  $CVaR_{0.95}$  of unmet demand by 85%. Planners can buy resilience in measurable units.
2. **Breadth beats depth.** The risk-averse policy opens two extra hubs (including a coastal site) rather than upsizing existing ones. Network breadth wins once a hurricane can isolate a region.
3. **Aggregate coverage hides commodity risk.** Water can be 100% covered while medicine collapses. Reporting must be per commodity.
4. **Early-period response matters most.** The risk-averse policy delivers 76.5% of demand in period one of the worst hurricane, versus below 1% for the baseline.

# 14. Conclusion and Outlook

## What this paper adds

- ▶ A two-stage stochastic MILP with mean-CVaR on unmet demand.
- ▶ Two transport modes with mode-specific delivery delays.
- ▶ Validation on the Rawls Gulf Coast case study, 15 real hurricanes.
- ▶ An efficiency and resilience Pareto curve that planners can read directly.

## Outlook

- ▶ Richer real hurricane data and refined commodity economics.
- ▶ Dynamic re-routing in the spirit of Yi and Özdamar.
- ▶ Equity constraints that bound the worst-off region's coverage.

## Take-home

Risk-averse humanitarian planning is not free, but its price is quantifiable and its benefit is delivered exactly where it matters, in the tail.

## 15. References

- [1] Rawls, C. G., and Turnquist, M. A. (2010). Pre-positioning of emergency supplies for disaster response. *Transportation Research Part B: Methodological*, 44(4), 521–534.
- [2] Yi, W., and Özdamar, L. (2007). A dynamic logistics coordination model for evacuation and support in disaster response activities. *European Journal of Operational Research*, 179(3), 1177–1193.
- [3] Noyan, N. (2012). Risk-averse two-stage stochastic programming with an application to disaster management. *Computers & Operations Research*, 39(3), 541–559.
- [4] Rockafellar, R. T., and Uryasev, S. (2000). Optimization of conditional value-at-risk. *Journal of Risk*, 2, 21–41.

# Thank you

Questions and discussion welcome.

Repository. <https://github.com/PNeDIT/mgthn0197-humanitarian-cvar>